

Understanding Capacity & Sensory Regulation

The research behind My Brain Companion: why capacity fluctuates, why sensory regulation matters, and what the evidence says helps.

Sensory Differences Are Common, Not Rare

Sensory processing and integration differences are estimated to affect somewhere between 5 and 25 per cent of children generally, with prevalence far higher in autistic children specifically. Estimates of sensory symptoms among autistic people range from roughly 30 per cent up to over 96 per cent across studies, depending on how sensory differences are measured (Dawson & Watling, 2000; Marco et al., 2011; Tavassoli et al., 2016, 2018). Sensory processing differences are considered a core, diagnostic feature of autism, not a secondary or occasional trait. They've been part of clinical descriptions of autism since its earliest documentation.

In practice, this means a young person may be hypersensitive (over-reactive) or hyposensitive (under-reactive) to sensory input, or experience it as fragmented or hard to filter. The mismatch between what an environment demands and what a person's nervous system can actually process is what drives difficulty engaging, not a lack of willingness (Tomchek & Dunn, 2007).

Why Capacity Fluctuates

Sensory differences don't just affect comfort. They measurably affect participation. Research links sensory processing differences in autistic children to differences in classroom participation, mealtimes, sleep, and social engagement, and in autistic adults, to participation in higher education and daily life more broadly (multiple studies reviewed in Peters & Marriott, 2022). This is the evidence base behind a premise My Brain Companion is built on: capacity isn't fixed day to day, and planning around the capacity someone actually has, rather than the capacity a schedule assumes they have, is a reasonable, evidence-consistent response, not an excuse.

Why this matters for Brain Before the Day

The journal's core question, 'how much of myself do I have available today?', isn't a soft framing device. It reflects a well-documented reality: sensory load and capacity genuinely vary, and naming that variation is the first step in working with it rather than against it.

Does Sensory Integration Therapy Actually Work?

The evidence is genuinely mixed, and it's worth being honest about that rather than overselling it. A 2015 controlled study of 34 autistic children (aged 40–65 months) found that six months of sensory integration sessions, three times a week, produced significant improvements in both gross and fine motor skills, measured using the Peabody Developmental Motor Scale (Abdel Karim & Mohammed, 2015). That result sits alongside a broader, mixed picture: systematic reviews report consistent improvement in motor skills specifically, but more inconsistent evidence for other outcomes like sensory difficulties or broader behaviour (medRxiv systematic review, 2024). One review found strong evidence for individually generated participation goals, but only “emerging” evidence for play, language, and social skills (Schaaf et al., 2018). A UK-funded randomised controlled trial (SenITA) was commissioned specifically because national guidelines concluded more research was still needed (Randell et al., 2022).

The honest summary: sensory-based approaches are widely used and show real promise, especially for motor and individually-defined participation goals, but they're not a guaranteed fix for every difficulty, and claims beyond what the evidence supports should be avoided.

How This Shapes MettleNest's Approach

This evidence base supports three things already built into MettleNest's neurodivergent-focused resources:

First, treating sensory and capacity differences as expected and common, not as a rare complication. The prevalence data alone justifies that framing. Second, building tools that help someone notice and name their own sensory and energy state, rather than tools that try to override or ignore it. Third, staying honest about what's proven versus promising: sensory-based strategies are worth offering as one part of a support toolkit, not marketed as a stand-alone cure.

References

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